

Introduction to Sampling Theory

Lecture 15 Ratio Method of Estimation



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Slides can be downloaded from
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Ratio Method of Estimation:

An important objective in any statistical estimation procedure is to obtain the estimators of parameters of interest with more precision.

It is also well understood that incorporation of more information in the estimation procedure yields better estimators, provided the information is valid and proper.

Use of such auxiliary information is made through the ratio method of estimation to obtain an improved estimator of population mean.

Ratio Method of Estimation:

Ratio Method of Estimation:

In ratio method of estimation, auxiliary information on a variable is available, which is linearly related to the variable under study and is utilized to estimate the population mean.

Let Y be the variable under study and X be any auxiliary variable which is correlated with Y .

The observation x_i on X and y_i on Y are obtained for each sampling unit.

Ratio Method of Estimation:

The population mean $\bar{X}$ of X (or equivalently the population total X_{tot}) must be known.

For example, x_i 's may be the values of y_i 's from

- some earlier completed census,
- some earlier surveys,
- some characteristic on which it is easy to obtain information etc.

For example, if y_i is the quantity of fruits produced in the i^{th} plot, then x_i can be the area of i^{th} plot or the production of fruit in the same plot in previous year.

Ratio Method of Estimation:

Let $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be the random sample of size n on paired variable (X, Y) drawn, preferably by SRSWOR, from a population of size N . The ratio estimate of population mean $\bar{Y}$ is

$$\hat{\bar{Y}}_R = \frac{\bar{y}}{\bar{x}} \bar{X} = \hat{R} \bar{X}$$

assuming the population mean $\bar{X}$ is known.

Ratio Method of Estimation:

The ratio estimator of population total $Y_{tot} = \sum_{i=1}^N Y_i$ is

$$\hat{Y}_{R(tot)} = \frac{y_{tot}}{x_{tot}} X_{tot}$$

where $X_{tot} = \sum_{i=1}^N X_i$ is the population total of X which is assumed to be known, $y_{tot} = \sum_{i=1}^n y_i$ and $x_{tot} = \sum_{i=1}^n x_i$ are the sample totals of Y and X respectively.

The $\hat{Y}_{R(tot)}$ can be equivalently expressed as

$$\begin{aligned}\hat{Y}_{R(tot)} &= \frac{\bar{y}}{\bar{x}} X_{tot} \\ &= \hat{R} X_{tot}.\end{aligned}$$

Ratio Method of Estimation:

Looking at the structure of ratio estimators, note that the ratio method estimates the relative change $\frac{Y_{tot}}{X_{tot}}$ that occurred after (x_i, y_i) were observed.

It is clear that if the variation among the values of $\frac{y_i}{x_i}$ is nearly same for all $i = 1, 2, \dots, n$ then values of $\frac{\bar{y}}{\bar{x}}$ (or equivalently $\frac{y_{tot}}{x_{tot}}$) vary little from sample to sample and the ratio estimate will be of high precision.

Bias and Mean Squared Error of Ratio Estimator:

Assume that the random sample $(x_i, y_i), i = 1, 2, \dots, n$ is drawn by SRSWOR and population mean $\bar{X}$ is known. Then

$$E(\hat{Y}_R) = \frac{1}{\binom{N}{n}} \sum_{i=1}^{\binom{N}{n}} \frac{\bar{y}_i}{\bar{x}_i} \bar{X}$$

$\neq \bar{Y}$ (in general).

Moreover, it is difficult to find the exact expression for $E\left(\frac{\bar{y}}{\bar{x}}\right)$ and $E\left(\frac{\bar{y}^2}{\bar{x}^2}\right)$. So we approximate them.

Bias and Mean Squared Error of Ratio Estimator:

We approximate the bias and mean squared error (mse) as follows:

Let

$$\varepsilon_0 = \frac{\bar{y} - \bar{Y}}{\bar{Y}} \Rightarrow \bar{y} = (1 + \varepsilon_0)\bar{Y}$$

$$\varepsilon_1 = \frac{\bar{x} - \bar{X}}{\bar{X}} \Rightarrow \bar{x} = (1 + \varepsilon_1)\bar{X}.$$

Since SRSWOR is being followed, so

$$E(\varepsilon_0) = 0$$

$$E(\varepsilon_1) = 0$$

Bias and Mean Squared Error of Ratio Estimator:

$$\begin{aligned} E(\varepsilon_0^2) &= \frac{1}{\bar{Y}^2} E(\bar{y} - \bar{Y})^2 \\ &= \frac{1}{\bar{Y}^2} \frac{N-n}{Nn} S_Y^2 \\ &= \frac{f}{n} \frac{S_Y^2}{\bar{Y}^2} \\ &= \frac{f}{n} C_Y^2 \end{aligned}$$

where $f = \frac{N-n}{N}$, $S_Y^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i - \bar{Y})^2$ and $C_Y = \frac{S_Y}{\bar{Y}}$

is the coefficient of variation related to Y .

Bias and Mean Squared Error of Ratio Estimator:

Similarly $E(\varepsilon_1^2) = \frac{f}{n} C_X^2$

$$\begin{aligned} E(\varepsilon_0 \varepsilon_1) &= \frac{1}{\bar{X}\bar{Y}} E[(\bar{x} - \bar{X})(\bar{y} - \bar{Y})] \\ &= \frac{1}{\bar{X}\bar{Y}} \frac{N-n}{Nn} \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y}) \\ &= \frac{1}{\bar{X}\bar{Y}} \frac{f}{n} S_{XY} \\ &= \frac{1}{\bar{X}\bar{Y}} \frac{f}{n} \rho S_X S_Y \\ &= \frac{f}{n} \rho \frac{S_X}{\bar{X}} \frac{S_Y}{\bar{Y}} \\ &= \frac{f}{n} \rho C_X C_Y \end{aligned}$$

where $C_X = \frac{S_X}{\bar{X}}$ is the coefficient of variation related to X and ρ is the population correlation coefficient between X and Y .